

Yin (Carry) & Yang (Convexity) Require Each Other to Survive & Thrive

A “carry trade” is a fairly universal term and concept across financial markets. But what does this label actually mean? An asset deemed to have “positive carry” is an asset that generates income completely independent of underlying price appreciation for the holders of that asset. A “positive carry” asset, simply put, is one which produces positive returns when the price of the asset does not move at all. As examples, equities holders can collect dividend yield and physical commodity owners can reap in the benefits of a convenience yield.

Does the logic behind a carry trade sound familiar for the onchain-native readers just yet? If not, it should. The word “yield” is one of the most common buzzwords used throughout all of DeFi and also happens to be the operative word for both the equities-centric and commodities-centric examples that were used in the preceding paragraph. What fundamental building block of DeFi can output yield while token prices do nothing? An oldie but a goodie: LPing on @Uniswap.

Carry traders & LPs have a psychological need for consistent gains...often at a severe eventual price.

Carry traders in all asset classes and Uniswap LPs are bound by a common desire for small, steady returns. In a nutshell, they share a mutual preference for negative return skewness where persistently marginal revenues can be quickly wiped out (and then some) in the face of excess volatility. There is a conscious tradeoff made with accruing this form of regular income in that carry traders also assume the risk of their portfolios blowing up with a single adverse market event. What would incline carry traders and LPs to prefer this type of return skewness profile?

There are intertwined and well-established phenomena within behavioral finance/economics known as prospect theory and the disposition effect, which essentially stipulate that investors are diminishingly receptive to gains and increasingly receptive to losses. Let's consider two supporting examples.

A trader has picked a winning stock with a 30% unrealized gain, but there's still a 50% chance that the stock picks up further steam tomorrow for a 60% total

gain. The trader will likely cash in by selling the stock now and guaranteeing the 30% in upfront income, thereby forgoing future potential profits.

A trader has picked a losing stock with a 30% unrealized loss, but there's a 50% chance that the losses will steepen even further to a 60% unrealized loss tomorrow. The trader will probably roll the dice and refrain from selling the stock in the hope that the unrealized losses can eventually be recouped.

Carry traders and Uniswap LPs fall within this exact mold. They will endure the prospect of incurring major losses for the sturdy guarantee of upfront yield and income. They are actively choosing to embrace the left tail of the return distribution (negative skewness) in exchange for a reliable revenue stream.

Changing of the Guard?

One of the most common carry trades to exploit among TradFi volatility markets is signified by the trait that implied volatility (IV) often trades at a premium to realized volatility (RV), which is monetized by selling options. The below figure tracks the price history of the CBOE S&P 500 One-Week PutWrite Index (WPUT) from 2017 through 2020, an indice that mimics the returns of a cash-secured put strategy on the S&P 500. Notice how the index displays a “sawtooth” pattern of gradual rising being rapidly extinguished with the market turmoil of “Volmageddon” in 2018 or the onset of the COVID-19 crisis in early 2020. This is a prime exemplification of the fragility and brittleness of negative return skewness in action.

S&P 500 options sellers are not only traditionally rewarded for IV being expensive to RV, but they are also accordingly betting that underlying prices will be range-bound or mean-reverting during a certain window. For hedging purposes, options sellers will be forced to buy into rallies and sell into downturns to maintain directional neutrality. Any extended price trends (either upwards or downwards) will therefore induce options sellers to chase the market at an accelerated rate with losses being worsened in the process. This is the exact type of trendy behavior that was episodically exhibited by the S&P 500 between 2017 and 2020, which translates to an even more hazardous environment for harvesting volatility carry. This 3-year timeframe (2017-2020) is representative of some relatively unique dynamics for TradFi equities volatility trading, but what if we told you that this TradFi abnormality was the pre-existing norm for onchain volatility markets?

Panoptic has long publicized how we are effectively completing the market for Uniswap LP shares by allowing them to be shorted (convexity to be bought). However, as LP positions constitute short perpetual put options, this onslaught

of options selling applies constant downward pressure upon onchain IV. Consequently, with the recent RV-IV differential in the ETH/USDC v3 5bps pool serving as the below example, IV trades cheaply to RV in our world.

Coupling this flipped RV-IV interchange with the empirical reality that crypto is likely the most momentum-prone and fat-tailed of all asset classes results in a climate where there is a natural incentive to purchase optionality rather than selling it. Panoptic is the platform that is packaging this trade for the masses. We are first-movers in bringing “antifragility” onchain.

Bizarro World Behavior

The question then becomes in this previously described bizarro world of volatility trading: what are the statistical traits that come with these inverted behavioral incentives? One of the defining characteristics for this new paradigm would be seeking out positive (as opposed to negative) return skewness. Positive return skewness is marked by incrementally incurring losses for the eventual probability of a major payout over an abbreviated period. Preference for positive (negative) return skewness can hence be likened to a desire for purchasing lottery tickets (selling insurance).

As a byproduct of these altered payoff profile biases, conventional prospect theory is also flipped on its head. Buyers of optionality/convexity are inclined to ride out winners for as long as possible, while simultaneously cutting losers as early as possible. This “inverse prospect theory” projects investors as being motivated in trying to compound gains and immediately undercut losses.

Note in the below graph how the resident convexity-buying strategy on Panoptic (the Unicorn Vault) forms a “stairstep” pattern. This type of trade payoff supplies a stark departure from the sawtooth pattern that is typically produced by a carry trade. Predominant periods of inactivity are intermixed with sudden and sizable gains.

On an elemental basis, what the Unicorn Vault represents is an “anti-carry” trade. All observed norms in TradFi volatility markets are flipped on their respective axes. But as we’ve demonstrated here, the playing field is already inherently tilted towards onchain “anti-carry” traders. We are looking to level the playing field for onchain carry traders (LPs) as well. The Uniswap v4 fee switch has provided us with a premier opportunity to do so.

The Onchain Trading Jungle

Uniswap LPs were already being undercompensated as a downstream effect of

the onchain “ $RV > IV$ ” precedent that we have highlighted. Yet, with the recent enactment of the Uniswap v4 fee switch, LP revenues are mechanically declining to an even greater degree. The onchain volatility carry trade (LPing) is no longer remotely viable on Uniswap.

Through a platform-native spread multiplier mechanism, LPs can earn up to 20% more in fees by staking their Uniswap positions on Panoptic with our newly-debuted LP migration UI. Over just around a 5-day window, the below sample trade reveals how a staked Uniswap LP position on Panoptic comparatively earned ~40% more in fees within the ETH/USDC v3 5bps pool as a beneficiary of the spread multiplier. The spread multiplier also continues to expand with increased options-buying demand, and considering the abundance of cheap onchain convexity that we’ve unlocked, options sellers can expect to be well-remunerated for their services.

Earning 20% more in fees is simply the beginning of the “Panoptic-LPing-as-option-selling.” It will take time to potentially phase onchain volatility markets into a “new normal” equilibrium. After all, these markets were structurally incomplete until Panoptic came online. Buying convexity was not logistically possible.

Although, once this pervasive IV compression has been corrected, the LPing could be on completely equal footing with onchain convexity purchasing. Onchain volatility markets could start to take on more TradFi-esque attributes with IV being greater than RV. But until that inflection point is reached, a 20% boost in prospective yield should come as a welcome alternative for Uniswap LPs.

Organizationally, trading marketplaces are not all that different from complex biological ecosystems or animal kingdoms. Every individual organism (trader type) has an important niche to occupy. Every participant has to discern where it fits among the broader hierarchy.

Carry traders require convexity seekers (anti-carry traders) to survive and vice versa. It is a symbiotic relationship. One hand washes the other until some form of stable equilibrium (or homeostasis) is attained, which in turn, will hopefully give way to long-term ecological evolution.

We at Panoptic are not attempting to disrupt this overarching balance. On the contrary, we are trying to introduce it. There will always be traders who have personal utilities/preferences skewed towards selling insurance to collect upfront premium, and conversely, there will always be those who willingly pay a premium for the privilege of waiting on a possible jackpot payday.

We are simply allotting these two polarized groups along the volatility trading continuum with the liberty to interact. In the coming weeks, we will be honing in on each segment of this convexity spectrum to expound on how Panoptic all-encompassingly improves and enlarges the full gamut of the onchain optionality offering. Have you figured out where you fit among the onchain trading jungle yet? If so, great! Panoptic is awaiting to evolve right alongside you.

Join the growing community of Panoptimists and be the first to hear our latest updates by following us on our social media platforms. To learn more about Panoptic and all things DeFi options, check out our docs and head to our website.